EC1410 – Fall 2026

Midterm Solutions
8:40-9:40am, November 2, 2026
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1. Across US metropolitan areas, Glaeser and Gottlieb (2009) estimate the following relationship between MSA population and MSA output per person (GMP per capita),

$$\ln(\text{GMP}) = 0.13 \times \ln(\text{Pop.}) + 8.8, \qquad R^2 = 0.25,\ N = 363.$$

   (a) *(5) Suppose that this estimation correctly describes the relationship between MSA size and output per person. Suppose City 1 has $Pop_1 = N$ and $GMP_1 = 1$. If City 2 has $Pop_2 = 2N$, what is the expected value of GMP_2?*

   Solution. Taking the difference between the two cities,

$$\ln(\text{GMP}_2) - \ln(\text{GMP}_1) = 0.13\big(\ln(2N) - \ln(N)\big) = 0.13 \ln 2,$$
$$\frac{\text{GMP}_2}{\text{GMP}_1} = 2^{0.13} \approx 1.094.$$

   Since $GMP_1 = 1$, the expected value is $GMP_2 \approx 1.09$; doubling population is associated with about a 9.4% increase in output per person.

   (b) *(5) Should we conclude from this estimate that larger cities cause increases in output per person? Explain briefly.*

   Solution. No. The estimate is a cross-sectional correlation and cannot by itself establish causation. Larger cities could have higher output per person because of *sorting* (more-productive workers and firms choose big cities) or *natural advantage* (inherently productive places grew large), rather than because size itself raises productivity.

2. Consider the monocentric city model with housing, an open city, and perfectly competitive housing production. Let $\overline{u} = 2$. The household's problem is

$$\max_{c,h,x} c^{1/2} h^{1/2}$$
$$\text{s.t. } w = c + ph + 2tx.$$

   Letting $\widetilde{w} = w - 2tx$, the household solves

$$\max_{c,h} c^{1/2} h^{1/2}$$
$$\text{s.t. } \widetilde{w} = c + ph.$$

   (a) *(10) Find the demand for housing h^* and consumption c^* in terms of p and $\widetilde{w}$.*

   Solution. Substituting $c = \widetilde{w} - ph$, the household maximizes

$$(\widetilde{w} - ph)^{1/2} h^{1/2}.$$

The first-order condition with respect to h gives

$$\begin{aligned} -\tfrac{1}{2}p(\widetilde{w}-ph)^{-1/2}h^{1/2}+\tfrac{1}{2}(\widetilde{w}-ph)^{1/2}h^{-1/2}&=0\\ \implies \quad ph&=\widetilde{w}-ph\\ \implies \quad h^*&=\frac{\widetilde{w}}{2p}. \end{aligned}$$

Substituting back into the constraint gives consumption,

$$c^*=\widetilde{w}-ph^*=\frac{\widetilde{w}}{2}.$$

(b) *(10) Use the fact that utility equals $\overline{u}=2$ everywhere to solve for the housing price p^* in terms of $\widetilde{w}$.*

Solution. In an open-city equilibrium utility is $\overline{u}=2$ everywhere, so

$$\overline{u}=(c^*)^{1/2}(h^*)^{1/2}=\left(\tfrac{\widetilde{w}}{2}\right)^{1/2}\left(\tfrac{\widetilde{w}}{2p}\right)^{1/2}=\frac{\widetilde{w}}{2\sqrt{p}},$$

$$2=\frac{\widetilde{w}}{2\sqrt{p^*}} \implies p^*=\frac{\widetilde{w}^2}{16}.$$

(c) *(10) Substitute $\widetilde{w}=w-2tx$ to write the housing price gradient p^* in terms of w, t, and x.*

Solution. Substituting $\widetilde{w}=w-2tx$,

$$p^*(x)=\frac{(w-2tx)^2}{16}.$$

3. This problem examines the rent and sale-price gradients during the COVID pandemic, as estimated by Gupta et al. (2021). Before the pandemic,

$$\ln R_0(x)=7.6-0.04\ln(x+1), \qquad \ln P_0(x)=13.2-0.127\ln(x+1),$$

and during the pandemic,

$$\ln R_1(x)=7.5-0.004\ln(x+1), \qquad \ln P_1(x)=13.15-0.115\ln(x+1),$$

where R is monthly rent, P is sale price, and x is distance (km) from the center.

(a) *(10) Find the percent change in rent at $x=0$.*

Solution. At $x=0$ the $\ln(x+1)$ terms vanish, so $\ln R_0(0)=7.6$ and $\ln R_1(0)=7.5$. The percent change in rent is

$$\frac{R_1(0)}{R_0(0)}=e^{7.5-7.6}=e^{-0.1}\approx 0.905,$$

a change of about -9.5%. (In levels, $R_0(0)=e^{7.6}\approx 1998$ and $R_1(0)=e^{7.5}\approx 1808$ per month.)

(b) *(10) Find the percent change in sale price at $x = 0$.*

Solution. Likewise, the percent change in the sale price is

$$\frac{P_1(0)}{P_0(0)} = e^{13.15-13.2} = e^{-0.05} \approx 0.951,$$

about -4.9%.

(c) *(10) Suppose the pandemic change in rent at $x = 0$ was permanent. Let ρ denote the interest rate. Using the rule for capitalization of rents into asset prices that we derived in class, we have $P = R/\rho$. If this relationship holds in practice, what percent change in the asset price does this imply for houses at $x = 0$?*

Solution. Since $P = R/\rho$, the price is proportional to the rent, so a *permanent* rent change produces an equal percent change in price,

$$\%\Delta P = \%\Delta R \approx -9.5\%.$$

(d) *(10) You now have two estimates of how asset prices at $x = 0$ changed: the actual change from (b) and the implied-if-permanent change from (c). What does comparing these two estimates suggest about how long people expected the rent change to last?*

Solution. The implied-if-permanent price drop ($\approx -9.5\%$) is about twice the size of the drop actually observed ($\approx -4.9\%$). One explanation for the fact that asset prices fell less than a permanent rent decline requires is that buyers and sellers seem to have expected the central-rent decline to be temporary.

4. This problem analyzes exactly the situation studied by Sandra Black (1999): a school attendance-zone boundary across which school quality jumps, and how that jump is reflected in land prices. In an open, linear monocentric city without housing, utility is

$$u(A(x)c) = \ln(A(x)c - 1),$$

with $\overline{u} = 0$, $\overline{R} = 0$, $\bar{l} = 1$, $p_c = 1$, $w = 6$, and $t = \frac{1}{2}$. A boundary sits at $x^* = 2$, where school quality jumps up:

$$A(x) = \begin{cases} A_0 = 1 & 0 \le x < x^*, \\ A_1 = 2 & x \ge x^*. \end{cases}$$

(a) *(6) Solve for and graph the equilibrium rent gradient $R(x)$. Note the discontinuity at x^*.*

Solution. With $\bar{l} = 1$ and $p_c = 1$, the budget gives

$$R(x) = w - c(x) - 2t|x| = 6 - c(x) - |x|,$$

so

$$R(x) = \begin{cases} 6 - 2 - x = 4 - x & 0 \le x < 2, \\ 6 - 1 - x = 5 - x & 2 \le x \le \overline{x}, \\ \overline{R} = 0 & x > \overline{x}. \end{cases}$$

The edge solves $5 - \overline{x} = 0$, so $\overline{x} = 5$. The gradient starts at $R(0) = 4$, falls to $R(2^-) = 2$, *jumps up* to $R(2^+) = 3$, then falls to $R(5) = 0$ (symmetric for $x < 0$): two segments of slope -1 with an upward step of height 1 at $x = 2$.

(b) *(4) Find equilibrium consumption $c(x)$ on each side of the boundary. By how much does consumption fall as a household crosses from just inside x^* to just outside?*

Solution. In an open-city equilibrium utility equals the reservation level everywhere,

$$u(A(x)c(x)) = \overline{u} = 0.$$

Using the functional form for u,

$$\ln(A(x)c(x) - 1) = 0 \implies A(x)c(x) = 2 \implies c(x) = \frac{2}{A(x)}.$$

It follows that $c(x) = 2$ for $0 \le x < x^*$ (where $A_0 = 1$) and $c(x) = 1$ for $x \ge x^*$ (where $A_1 = 2$). Crossing from just inside to just outside x^*, consumption falls by $2 - 1 = 1$. With better schools, households reach $\overline{u}$ while consuming less.

(c) *(5) Show that the size of the jump in the rent gradient at x^* is exactly equal to the decline in consumption at x^* that you found in part (b).*

Solution. Let $\varepsilon > 0$ be a small distance. The jump in rent across the boundary is

$$\begin{aligned} R(x^* + \varepsilon) - R(x^* - \varepsilon) &= \big[w - c_{\text{out}} - 2t(x^* + \varepsilon)\big] - \big[w - c_{\text{in}} - 2t(x^* - \varepsilon)\big] \\ &= (c_{\text{in}} - c_{\text{out}}) - 4t\varepsilon \\ &\approx c_{\text{in}} - c_{\text{out}} = 2 - 1 = 1, \end{aligned}$$

exactly the consumption decline calculated in (b).

(d) *(5) Black (1999) values school quality by comparing house prices for homes just on either side of such a boundary. Using your answer to (c), explain why the price jump at the boundary measures households' willingness to pay for the better schools, and why the fact that commuting distance is essentially the same on both sides is what makes this "border discontinuity" comparison valid.*

Solution. Free mobility requires that utility be the same on both sides of the school district boundary. Therefore, a household just outside gives up exactly $c_{\text{in}} - c_{\text{out}}$ in consumption in exchange for the better schools; since commuting is unchanged across the boundary, that forgone consumption is paid entirely as higher rent. Thus the rent (hence price) jump equals households' willingness to pay for the school-quality difference. This is why Black restricts attention to transactions within a few hundred yards of a boundary: holding commute distance and other location factors fixed, the price discontinuity isolates the value of school quality.